PHYSICAL CHEMISTRY CUMULATIVE EXAM

April 23, 1994

I. The motion of an object in the +z direction is opposed by a force $F_z = az + b$, where a & b are constants:

a) Derive a general expression for the work done in traversing the distance from z_1 to z_2

b) Find the work done in moving the object through a distance of 10.0 meters, starting at

1) $z_1 = 0.0\text{m}$

2) $z_1 = 20.0\text{m}$

where $a = 300\ \text{N}\cdot\text{m}^{-1}$ and $b = 500\ \text{N}$

II. The motion of an object of mass m along the +x-axis is opposed by a force $F = b \sin(\pi x/a)$, where b & a are constants.

a) Derive a general expression for the work needed to move the object from x_1 to x_2

b) Calculate the work needed to move the object from $x_1 = 0$ to

1) $x_2 = a$

2) $x_2 = 2a$

III. A particular gas obeys the virial equation of state

$$pV_m/RT = 1 + B/V_m + C/V_m^2 + \ldots$$

where p is the pressure, V_m is the molar volume, T is the temperature, R is the gas constant, and B,C, etc. are the virial coefficients.

a) Derive a general expression for the work done when the gas volume changes isothermally and reversibly from V_1 to V_2

b) How does this compare to the work done in the case of an ideal (i.e., perfect) gas?

c) For argon, $B = -21.7\text{cm}^3\cdot\text{mol}^{-1}$ and $C = 1200\ \text{cm}^6\cdot\text{mol}^{-2}$ at $T = 273\text{K}$. Calculate the work done when 1.00 mol of argon expands from 500 to 1000 cm^3

$\{R = 8.314\ \text{J}\cdot\text{K}^{-1}\text{mol}^{-1} = 0.08206\ \text{L}\cdot\text{atm}\cdot\text{K}^{-1}\cdot\text{mol}^{-1}\}$